26ECCA0	ELECTRIC AND MAGNETIC CIRCUITS	Category	L	T	P	Credits
		PCC	2	1	0	**3**

Preamble

This course introduces basic electrical circuit concepts, including voltage, current, power, and Kirchhoff's laws. It covers DC and AC circuit analysis using network theorems and phasor techniques, along with time-domain analysis of first-order RL and RC circuits. The course also includes resonance and an introduction to magnetically coupled circuits and transformers.

Prerequisite

NIL

Course Outcomes

After the successful completion of the course, students will be able to:

No.	Course Outcome	TPS Level	Performance Indicator	CO Weightage	SDG addressed	SDG Level
CO1	Apply fundamental electrical quantities and circuit laws (KCL, KVL) to DC circuits.	TPS3	1.2.1	10	SDG 9	2
CO2	Solve DC circuits using Node and Mesh analysis	TPS3	1.3.1	10	SDG 9	2
CO3	Apply network theorems to simplify electrical circuits.	TPS3	1.3.2	20	SDG 9	2
CO4	Demonstrate phasor representation and impedance concepts to AC circuits and compute power parameters.	TPS3	2.1.1	20	SDG 9	2
CO5	Apply transient analysis techniques to determine the response of RL and RC circuits with initial conditions.	TPS3	3.1.2	20	SDG 9	2
CO6	Apply resonance and coupled circuit concepts to analyze frequency response and magnetically coupled networks.	TPS3	5.1.1	20	SDG 9	2

Assessment Pattern

CO	CAT1	Assignment 1	CAT2	Assignment 2	Terminal Exam
	% of Marks	% of Marks	% of Marks	% of Marks	% of Marks
CO1	30	30			10
CO2	30	30			10
CO3	40	40			20
CO4			30	30	20
CO5			30	30	20
CO6			40	40	20
Total	**100**	**100**	**100**	**100**	**100**

Syllabus

Basic electrical quantities: Charge – current – voltage – power - Independent and dependent sources - Ohm's law - Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL) - Series and parallel circuits; Voltage and current divider rules.

Solution Methods: Node voltage method - Mesh current method - Supernode and supermesh analysis - Systematic formulation and solution of circuit equations for DC networks.

Network Theorems: Source transformation - Superposition theorem - Thévenin's and Norton's equivalent circuits - Maximum power transfer theorem - Δ–Y and Y–Δ transformations - Simplification of complex networks using theorems.

AC Analysis: Sinusoidal signals and phasor representation - Complex impedance of R, L and C elements - Analysis of RL, RC and RLC series and parallel circuits - Active, reactive and apparent power - Power factor - Basic introduction to three-phase systems.

Transient Analysis: Analysis of first-order RL and RC circuits using differential equation approach - Determination of initial and steady-state conditions - Natural and forced responses - Time constant and transient behavior under DC excitation - Evaluation of current and voltage responses

Resonance: Series and parallel resonance - Resonant frequency - Quality factor bandwidth and half-power frequencies

Magnetic Circuits: Mutual inductance and coefficient of coupling - Dot convention - Energy considerations - Basic analysis of coupled circuits - Ideal transformer and impedance matching (introductory level).

6. Learning Resources

6.1 Text Book

- William H. Hayt Jr., Jack E. Kemmerly, Steven M. Durbin, "Engineering Circuit Analysis", 10th Edition, McGraw Hill, 2025.
- Charles K. Alexander and Matthew N. O. Sadiku, "Fundamentals of Electric Circuits", 7th Edition, McGraw Hill, 2022.

6.2 Reference Books & Web resources

- Abhijit Chakrabarthi, Sudipta Debnath, chandan kumar chanda, "Basic Electrical Engineering", 2nd edition, Mcgraw Hill, 2021.
- James W. Nilsson and Susan Riedel, "Electric Circuits", 11th Edition, Pearson Education, 2019.
- A. Sudhakar and Shyammohan S. Palli, "Circuits and Networks: Analysis and Synthesis", 6th Edition, McGraw Hill, 2025.

NPTEL Courses

- A Basic Course on Electric and Magnetic Circuits; Prof. Ashok Kumar Pradhan; IIT Kharagpur
- Basic Electrical Circuits; By Prof. Gajendranath Chowdary|IIT Hyderabad
- Basic Electrical Circuits Nagendra Krishnapura; IIT Madras

7. Course Contents and Lecture Schedule

Module No	Topics	No of Periods	Course Outcomes
1	Basic electrical quantities: charge, current, voltage, power, Independent and dependent sources	1	
1	Ohm's law; Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL);	2	
1	Series and parallel circuits; Voltage and current divider rules	2	
2	Solution Methods: Node voltage method	1	
2	Mesh current method	1	
2	Supernode and supermesh analysis	2	
2	Systematic formulation and solution of circuit equations for DC networks	1	
3	Network Theorems: Source transformation; Superposition theorem	1	
3	Thévenin's and Norton's equivalent circuits	2	
3	Maximum power transfer theorem	1	
3	Δ–Y and Y–Δ transformations	1	
3	Simplification of complex networks using theorems	1	
4	AC Analysis: Sinusoidal signals and phasor representation; Complex impedance of R, L and C elements	2	
4	Analysis of RL, RC and RLC series and parallel circuits	2	
4	Active, reactive and apparent power; Power factor	1	
4	Basic introduction to three-phase systems	1	
5	Transient Analysis: Analysis of first-order RL and RC circuits using differential equation approach	2	
5	Determination of initial and steady-state conditions	2	
5	Natural and forced responses	1	
5	Time constant and transient behavior under DC excitation; Evaluation of current and voltage responses	2	
6	Resonance: Series and parallel resonance	1	
6	Resonant frequency; Quality factor bandwidth and half-power frequencies	2	
7	Magnetic Circuits: Mutual inductance and coefficient of coupling	2	
7	Dot convention; Energy considerations; Basic analysis of coupled circuits	1	
7	Ideal transformer and impedance matching (introductory level).	1	
Total Periods		**36**	

8. Course Designer(s):

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